Predicting Activation-Steering Failure from Construction-Sample Uncertainty

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Abstract

Activation steering provides a lightweight way to control the behavior of large language models (LLMs) at inference time by modifying internal activations without updating model parameters. However, a steering vector is typically estimated from a finite set of contrastive examples and then treated as fixed for every input. Recent work shows that steering effects can vary substantially across prompts and can even move in the opposite direction from the intended behavior [4, 5]. This project investigates whether failures can be predicted from uncertainty induced by the finite construction set itself. I propose *construction-sample uncertainty* (CSU), an input-conditioned measure obtained by bootstrap-resampling the examples used to construct a steering vector and measuring the variance of their projected steering effects on a new input. The central hypothesis is that CSU predicts per-input anti-steering failures beyond existing measures such as global vector consistency, steering projection, and model confidence. If supported, CSU can be used to selectively abstain from steering on inputs for which the intervention is unreliable, improving the risk–coverage trade-off. The project will evaluate this hypothesis on open instruction-tuned LLMs using controlled behavioral tasks, with a small open-ended safety evaluation as external validation.

1 Motivation and Research Gap

Activation steering modifies a model's hidden representations during inference to increase or decrease a desired behavior. Representation Engineering and related activation-intervention methods established that meaningful behavioral directions can be extracted from internal representations and manipulated without updating model parameters [1, 2]. Contrastive Activation Addition (CAA) subsequently provided a simple and effective procedure for constructing steering vectors by averaging activation differences between positive and negative examples [3].

Despite its simplicity, activation steering is not uniformly reliable. Recent studies show substantial variation in steering effects across prompts, concepts, models, and intervention configurations [4, 5, 6]. In particular, steering vectors may produce unintended or opposite effects for individual inputs even when their aggregate behavior appears successful [4, 5]. Existing analyses relate reliability to global properties of the construction data, such as cosine agreement among activation differences and separability of positive and negative activations [5]. Other work predicts steerability from early decoding states [6], and a further line of work abstains from intervening when correction is unlikely to succeed [7].

This project studies a narrower and complementary question: *does uncertainty caused by finite-sample estimation of the steering vector contain input-specific information about steering failure?* The key distinction is between uncertainty about the steering vector globally and uncertainty about its effect on a particular input. Two inputs can therefore receive the same globally estimated vector while having different uncertainty in the projected intervention effect.

2 Research Questions and Hypotheses

The primary research question is:

> **Does input-conditioned uncertainty from finite-sample steering-vector estimation predict per-input steering failure beyond existing measures of vector reliability, steering projection, and model confidence?**

The secondary question is whether this uncertainty can support a practical selective-steering policy: *can the model steer on reliable inputs while abstaining on unreliable ones to obtain a better risk–coverage trade-off?*

The project tests four hypotheses:

- **H1:** The projected effect of a steering vector varies measurably across bootstrap resamples of its construction set.

- **H2:** Input-conditioned CSU predicts mechanistic anti-steering failures.
- **H3:** CSU provides predictive information beyond steering projection, global vector reliability, construction-set size, and base-model confidence.
- **H4:** Ranking inputs by CSU enables a useful risk–coverage trade-off for selective steering.

3 Methodology

3.1 Activation Steering and Construction-Sample Uncertainty

Let $\mathcal{C} = \{(x_i^+, x_i^-)\}_{i=1}^n$ be a contrastive construction set and let $h(x) \in \mathbb{R}^d$ denote the residual-stream activation at a fixed intervention layer and token position. A CAA-style steering vector is

$$\bar{v} = \frac{1}{n} \sum_{i=1}^{n} \left[h(x_i^+) - h(x_i^-) \right]. \tag{1}$$

At inference time, the hidden state is modified as

$$h_{\text{steered}}(x) = h(x) + \alpha \bar{v}, \tag{2}$$

where α is the steering coefficient [3].

To quantify the effect of finite construction data, the construction set is bootstrap-resampled B times. For bootstrap sample b, a steering vector v_b is constructed using the same procedure. The empirical covariance is

$$\widehat{\Sigma}_v = \frac{1}{B-1} \sum_{b=1}^{B} (v_b - \bar{v})(v_b - \bar{v})^\top. \tag{3}$$

For a new input x, define *construction-sample uncertainty* as

$$U_{\text{CSU}}(x) = h(x)^\top \widehat{\Sigma}_v h(x) = \text{Var}_b \left[h(x)^\top v_b \right]. \tag{4}$$

Thus, CSU measures how much the estimated steering effect on *this particular input* changes when the construction examples are resampled. It is therefore distinct from a global measure of vector instability.

The main baselines will include absolute and signed steering projection, relative uncertainty

$$\frac{\sqrt{U_{\text{CSU}}(x)}}{|h(x)^\top \bar{v}| + \epsilon},$$

global cosine agreement among bootstrap vectors, positive/negative activation separability, construction-set size, vector norm, and base-model logit margin or confidence. The global cosine and separability baselines follow the dataset-level reliability predictors reported by Braun et al. [5].

3.2 Failure Definition

The primary analysis will use a controlled binary-choice setting in which the desired behavior can be represented by a target logit difference. Let

$$\Delta m(x) = m_{\text{steered}}(x) - m_{\text{base}}(x), \tag{5}$$

where m denotes the target logit difference. A *mechanistic anti-steering failure* occurs when $\Delta m(x) < 0$, meaning that the intervention moves the model opposite to its intended direction.

This endpoint is used as the main outcome because it isolates the effect of the intervention from downstream sampling variability. A smaller open-ended experiment will subsequently examine behavioral outcomes such as refusal or harmlessness as external validation.

3.3 Experimental Plan

The experiments form a sequence of increasingly demanding tests.

1. Establish the uncertainty signal. Bootstrap the construction set with $B \in \{5, 10, 20, 50, 100\}$ and measure the stability of U_{CSU}. The smallest sufficiently stable B will be used in the main study. Construction-set size will also be varied to test whether uncertainty decreases as more construction examples are available.

2. Test predictive value. For held-out inputs, compare nested predictive models:

1. steering projection only;
2. existing reliability measures;
3. CSU only;
4. projection plus existing reliability;
5. all baselines plus CSU.

The critical test is whether adding CSU improves out-of-sample prediction over the strongest baseline combination. Evaluation will use AUROC, average precision, calibration, and held-out performance.

3. Test whether CSU is input-specific. CSU will be compared with global uncertainty summaries. If global vector instability explains the result equally well, CSU does not provide the proposed additional information. Strong evidence for H3 requires that inputs with similar global reliability can nevertheless have different CSU values associated with different failure rates.

4. Test robustness. The relationship will be evaluated under prompt paraphrases, answer-order changes, alternative templates, and held-out behavioral examples. Construction and evaluation sets will remain disjoint to avoid leakage.

5. Evaluate selective steering. Inputs will be ranked by predicted reliability, and steering will be applied only to the selected fraction. Risk–coverage curves will compare CSU against random selection, steering projection, model confidence, global reliability, and a combined baseline. The goal is not to steer every input, but to achieve lower failure risk at a given coverage level.

6. Behavioral validation. A smaller open-ended evaluation will test whether low-CSU inputs also exhibit fewer undesirable steering outcomes. This experiment is external validation rather than the main scientific test.

4 Models, Data, and Feasibility

The primary implementation will use Qwen2.5-1.5B-Instruct for pipeline development and Qwen2.5-3B-Instruct for the main study. If resources permit, one additional small open model will test transfer across model families. Initial tasks will use controlled behavioral datasets suitable for binary evaluation, such as sycophancy or refusal-related behaviors, as used in CAA [3]. A small open-ended safety set will be used only after the mechanistic experiments.

The project requires no model fine-tuning. Activations can be cached and reused across bootstrap experiments, keeping computation manageable on limited GPU resources. The six-week plan is:

Week	Main objective
1	Implement CAA pipeline; select intervention layer and controlled task.
2	Build bootstrap procedure; validate CSU and activation caching.
3	Run predictive-value experiments; predefined go/no-go checkpoint.
4	Test input-specificity and robustness across prompt variations.
5	Evaluate selective steering and risk–coverage; run behavioral validation.
6	Consolidate results, analyze failure cases, prepare demo and final report.

5 Expected Contributions and Outcomes

The primary contribution is an empirical investigation of whether finite-sample uncertainty in steering-vector construction provides information about *which individual inputs are unsafe or ineffective to steer*. The proposed CSU quantity is a simple way to make this uncertainty input-conditioned. The scientific contribution, however, is not the covariance formula itself; it is the test of whether CSU adds predictive information beyond established reliability and input-geometry measures.

Three outcomes are possible. First, if CSU significantly improves failure prediction and selective steering, the project will provide evidence that construction-sample uncertainty is a useful reliability signal for activation steering. Second, if CSU helps only under particular construction-set sizes or behaviors, the result will characterize the regime in which the signal is useful. Third, if CSU adds little beyond simpler baselines, the project will provide a negative result showing that vector-estimation instability and per-input steerability may be largely

distinct phenomena. In all cases, the evaluation is designed to distinguish these explanations rather than assuming that the proposed measure must succeed.

The final demonstration will show the same prompts under three settings: unsteered, standard CAA, and CSU-aware selective steering. For each input, the system will display the predicted uncertainty, steering decision, and observed target-behavior change, together with an aggregate risk–coverage curve.

6 References

References

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